

Project: Knowledge Discovery from Data
CS4445 Data Mining and Knowledge Discovery in Databases - A Term 2026
Prof. Carolina Ruiz

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Important Note:

Having all of our code in a single file was infeasible. Instead we chose to have all of our relevant preprocessing set up inside the main file, which then loads the relevant data into .parquet files. Each of the subsequent files then use those .parquet files as the sources of these data, keeping any issues or consequences from one file from spilling into another. To run our programs, run the [main.ipynb](#) file, and then the file you wish to observe.

Part I. [50 points] Dataset Description, Exploration, and Preprocessing: (At most 2 pages)

Part I.1. [05 points] Dataset Description: (e.g., dataset domain, number of instances, number of attributes, distribution of target attribute, attributes that have missing values, ...)

The majority of the data are discrete values. There are 46 total data instances; 18 of them are discrete attributes with a qualitative value, and 2 of them are Discrete attributes with a numerical value. 14 of them are continuous numerical attributes, and 13 of them are attributes with only binary true/false values. The target attribute, severity, centers around 2. End Latitude and End Longitude are the most common offenders for missing data.

[10 points] Data Exploration: (e.g., comments on interesting or salient aspects of the dataset, visualizations, correlation, issues with the data, ...)

The Distance attribute is wildly skewed towards 0 miles, with only a couple of outliers that go beyond .01 miles. There are also a number of different pieces of data that seem to be encapsulating either entirely redundant, or irrelevant pieces of data. Reducing these extraneous pieces of information is very important for getting the most accurate predictions possible.

Part I.2. [05 points] Initial preprocessing, if any, based on data exploration findings: (e.g., removing IDs, strings, ...)

Note: Keep initial data preprocessing to a minimum - just the necessary - and instead experiment with data preprocessing in the following sections.

The dataset has 46 attributes. Several attributes are immediately irrelevant to our data mining effort, such as the ID attributes, the written Description attribute, and the Country attribute, which are either impossible to interpret or provide no value. This leaves us with 43 remaining attributes, including our target attribute for classification and regression techniques: Severity. Several of these attributes can be combined, such as condensing Street, City, County, State, and Zipcode into one, or potentially two, attributes. Furthermore, the beginning and ending Latitude and Longitude contribute the same redundant information. Zipcode provides the greatest location accuracy relative to the number of attributes required, but is very difficult to encode to a meaningful value. Time Zone and Airport code both are irrelevant, as they provide less specific, more general information. The Turning Loop attribute has 0 true values, so it is unhelpful. Sunrise_Sunset, along with the various twilight attributes, provides redundant information; however, we will use later supervised feature selection to eliminate it. Our advanced data preprocessing involved encoding all features into numeric values, plus feature selection and elimination to simplify the dataset. We started by copying our basic preprocessing so we could compare the results. We converted our start and end time values from datetime to integers, then created a new attribute for the duration in minutes. From that, we broke up the start time into hours, day of the

week, and month. We then dropped the start and end time values, since we had already extracted all relevant information. Next, latitude and longitude give us the relevant location information, meaning that the City, County, Zipcode, Street, Airport_Code, and State attributes are all redundant. Next, we could collate Sunrise_Sunset, Civil_Twilight, Nautical_Twilight, and Astronomical_Twilight. Each of these contains redundant information regarding the time of day, so we replaced them with a sunset attribute that reports if it is night or day. Next, we use one-hot encoding to turn wind direction and weather conditions into dense columns. We then drop Timezone and Weather_timestamp, as they are redundant. This leaves us with 187 total columns.

Part I.3. [30 points] Advanced data preprocessing: Describe data preprocessing and/or transformations you designed to make the dataset more informative and to help uncover more meaningful and useful car accident patterns.

Our advanced preprocessing techniques include identifying the road type, weather conditions, wind directions, wind hour, as well as other salient pieces of information. In addition, grouping various pieces of data relating the time into various attributes, encoding them to make it easier to analyze. This helped us reduce the complexity of the dataset significantly. In addition, we chose only the most relevant features by selecting the 14 more relevant attributes.

Part II. [135 points] Classification Models

Part II.1 Summary of Classification Experiments Using 10-fold Stratified Cross-validation

1. [10 points] What advanced data preprocessing did you apply to the training dataset for the classification experiments?

We used feature selection to choose only the most important features. We discovered that the most important features are: Duration, Distance, starting_lat and starting_long, pressure, lighting, day_of_week, and traffic_controlled

2. [60 points] Use the table below to record your results. Round results to 3 decimal places. Note: for the confusion matrix, enter just the 4x4 matrix values, not a visualization, to save space.

We put the results of each individual fold in a table that can be found here. Each of the different methods has a page of the table dedicated to it that details the confusion matrix, the accuracy, the precision, the recall, and the f1 score. In addition, due to time constraints when running the unlimited max_depth Decision Tree and Random Forest, we were forced to limit it.

<https://docs.google.com/spreadsheets/d/1VBII2gG4CPXSda3i1IqOvy-fg8cFiG-EepgcL8qbhk/edit?usp=sharing>

The data is also provided at the very end of this document, as a method

Dummy Classifier:

Accuracy mean: 0.933

Accuracy std: 0.000

Precision mean: 0.233

Precision std: 0.000

Recall mean: 0.250

Recall std: 0.000

F1 mean: 0.241

F1 std: 0.000

max_depth=None Decision Tree:

Accuracy mean: 0.930

Accuracy std: 0.000

Precision mean: 0.638

Precision std: 0.003

Recall mean: 0.646

Recall std: 0.002

F1 mean: 0.642

F1 std: 0.002

max_depth=None Decision Tree Classifier

Accuracy mean: 0.930

Accuracy std: 0.000

Precision mean: 0.638

Precision std: 0.003

Recall mean: 0.646

Recall std: 0.002

F1 mean: 0.642

F1 std: 0.002

Fold analysis:

max_depth=5 Decision Tree Classifier

Accuracy mean: 0.943

Accuracy std: 0.001

Precision mean: 0.706

Precision std: 0.012

Recall mean: 0.503

Recall std: 0.006

F1 mean: 0.555

F1 std: 0.006

max_depth=None Random Forest Classifier

Accuracy mean: 0.939

Accuracy std: 0.000

Precision mean: 0.654

Precision std: 0.018

Recall mean: 0.372

Recall std: 0.010

F1 mean: 0.420

F1 std: 0.011

max_depth=5 Random Forest Classifier

Accuracy mean: 0.934

Accuracy std: 0.000

Precision mean: 0.459

Precision std: 0.135

Recall mean: 0.256

Recall std: 0.005

F1 mean: 0.254

F1 std: 0.010

2. [10 points] Analyze the 10-fold stratified cross-validation results. Include any insightful observations. (At most ½ page)

To aid in the analysis, we have provided a table containing the results as observed above, with extra significant figures to aid in analysis.

	Model	Accuracy mean	Accuracy std	Precision mean	Precision std	Recall mean	Recall std	F1 mean	F1 std
0	Dummy	0.934045	0.000016	0.233511	0.000004	0.25	0	0.241474	0.000002
1	max_depth=None Decision Tree	0.940557	0.00149	0.663887	0.023921	0.486728	0.015163	0.533586	0.016281
2	max_depth=5 Decision Tree	0.936961	0.000533	0.670172	0.039749	0.38426	0.012378	0.400348	0.013346
3	max_depth=None Random Forest	0.938714	0.000399	0.653571	0.016446	0.370524	0.010721	0.418227	0.011849

4	max_depth=5 Random Forest	0.934192	0.000107	0.458546	0.075012	0.254957	0.003825	0.251104	0.007312
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As model complexity grows, more cross-validation seems to improve the results. The average accuracy increases slightly. The real growth comes in the increase in precision. Our dummy classifier has an average precision of .23; however, our decision trees with max_depth=5 and unlimited max_depth have more than double that precision, at .66. The best result comes from the Decision Tree with max_depth=5. Our other values are similarly good, but not as high as the accuracy or precision of the Decision Tree with max_depth=5. The Recall is better, though the decision tree with an unlimited max_depth is slightly higher. The average F1 score is also very good compared to the other methods; however, it is also beaten by the unlimited max_depth. Overall, the best option is undoubtedly the max_depth of 5. Across the metrics, it routinely scores as the best or one of the best and is therefore appropriate for our final classification model.

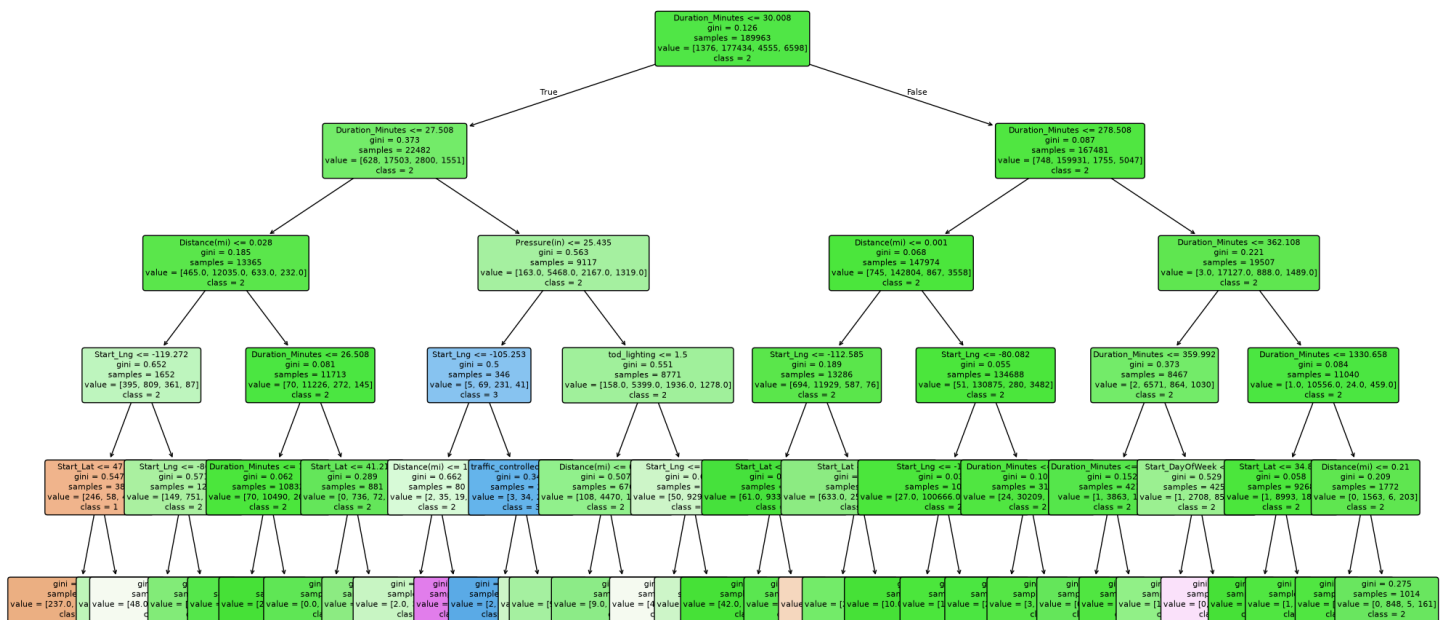
Part II.1 Final Classification Model and Analysis. (At most 2 pages)

- [10 Points] 3. What is the estimated classification accuracy of this decision tree over the entire domain of car accidents? Explain your answer.

Our estimated classification accuracy is the accuracy we already calculated. This estimated accuracy would be .937. This is because we have not yet used our more advanced preprocessing methods and are instead relying on a more basic dataset. As we build our final model by selecting only the relevant information, we can expect accuracy and other relevant metrics to increase.

- [5 Points] 4. Include a depiction of the decision tree in your report.

Final Decision Tree (max_depth=5)



- 5. Qualitative evaluation of the tree:

- [5 Points] How large is the tree?

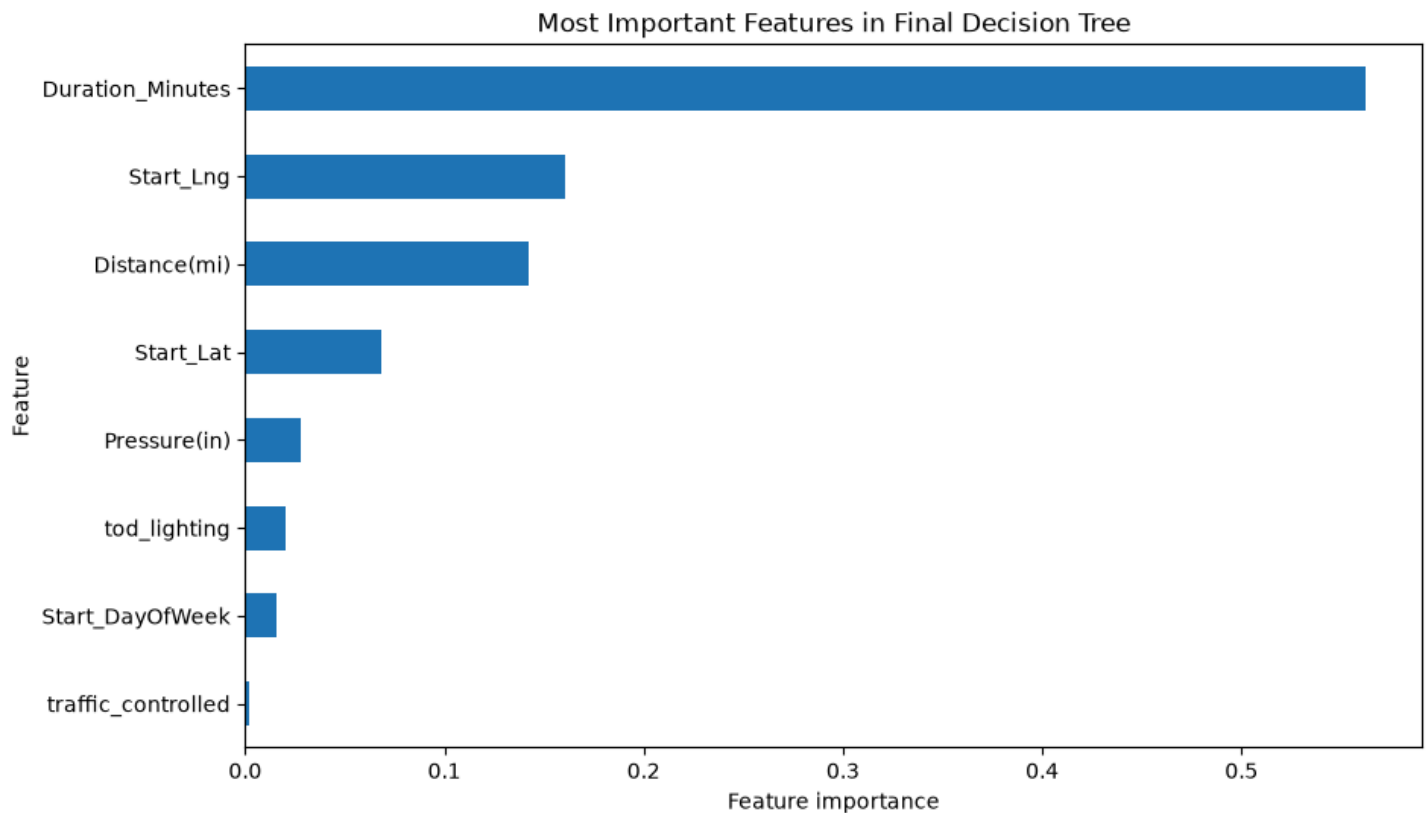
The Tree (obviously) has a maximum depth of 5. It splits into two possibilities at every level, so it has 32 leaves and 63 nodes.

- [5 Points] How readable is the tree?

The first couple of rows are rather readable, but once you get beyond the 3rd row, it begins to get rather difficult to read. This can be formatted, but otherwise, it is rather easy to follow the path needed to predict, given a dataset, using this tree.

- [15 points] Describe the 3 most salient car accident patterns uncovered by this tree and how they can be useful to domain experts to understand and/or prevent car accidents.

The most important feature we found is Duration in minutes. Accidents that last longer tend to be significantly more severe. Speculatively, a longer accident likely involves additional mitigating factors that prolong it, potentially injuring more people, being more deadly, or otherwise causing more damage to the vehicle or environment. The next of our most important features is the starting latitude. This also is consistent with our domain knowledge. As the latitude increases, you get closer to the northern poles, meaning that you are at a much greater risk for running into ice or other poor, wintry, conditions. Our third most important feature is the distance. The logic behind this feature is largely the same as the duration, as the longer a crash takes place over, the more likely it is to be deadly.



[15 points] Overall Analysis of Classification Results, Discussion, and/or Visualizations. Provide any additional overall analysis of the classification experiments and results.

The classification models needed some modifications to make them practical to run. The first thing is that instead of using the full dataset of 3700000 data points, we decided to reduce this to a 5% subset. This reduced the time needed to run our code from over an hour in total, to only around 10 minutes. The next adjustments to our program were in a similar vein. We chose to use a `max_depth=10` in place of `max_depth=None`. This increase should be sufficient to simulate an unlimited maximum depth, as by doubling our maximum depth, we increase the total possible leaves from 32 to 1024, and the number of leaves from

Part III. [135 points] Regression Models

Part III.1 Summary of Regression Experiments Using 10-fold Cross-validation

1. [10 points] What advanced data preprocessing did you apply to the training dataset for the regression experiments?

We aggregated and transformed certain features into new ones. For instance, Weather_Condition originally had many different categorical values that were different but had a lot of overlap. We mapped data instances into new categorical values by detecting keywords such as “snow” or “windy”. The rush_hour attribute was created to represent if an accident occurred during rush hour or not, and was made using the Start_Time attribute. The road_type attribute was created using the Street attribute, and represents either local or highway roads. For a complete overview of the preprocessing completed for regression, look at the table below.

Attribute created/transformed	Attributes used	Columns dropped after creation/transformation
Weather_Condition	Weather_Condition	N/A
rush_hour	Start_Hour	Start_Hour
tod_lighting	Sunrise_Sunset, Civil_Twilight	Sunrise_Sunset, Civil_Twilight, Nautical_Twilight, Astronautical_Twilight
traffic_controlled	Traffic_Signal, Stop, Give_Way	Traffic_Signal, Stop, Give_Way
junctions	Junction, Roundabout, No_Exit	Junction, Roundabout, No_Exit
traffic_calming	Bump, Traffic_Calming	Bump, Traffic_Calming
pedestrians	Crossing, Station, Amenity	Crossing, Station, Amenity
road_type	Street	Street

2. [60 points] Use the table below to record your results. Round results to 3 decimal points.

Cross-val Iterations	1	2	3	4	5	6	7	8	9	10	mean	std
Average Regressor												
Mean Squared Error (MSE)	15.688	16.676	16.071	16.632	15.535	16.696	15.667	16.717	16.598	16.537	16.282	0.462
Root Mean Squared Error (RMSE)	3.961	4.084	4.009	4.078	3.941	4.086	3.958	4.089	4.074	4.067	4.035	0.058
Mean Absolute Error (MAE)	1.742	1.791	1.756	1.781	1.732	1.792	1.742	1.790	1.784	1.780	1.769	0.022
R ²	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Linear Regressor												
Mean Squared Error (MSE)	15.361	16.330	15.715	16.218	15.199	16.302	15.308	16.356	16.233	16.167	15.919	0.447
Root Mean Squared Error (RMSE)	3.919	4.041	3.964	4.027	3.899	4.038	3.913	4.044	4.029	4.021	3.989	0.056
Mean Absolute Error (MAE)	1.709	1.759	1.730	1.752	1.701	1.762	1.707	1.759	1.752	1.744	1.737	0.023
R ²	0.021	0.021	0.022	0.025	0.021	0.024	0.023	0.022	0.022	0.022	0.022	0.001
Regression Tree (max_depth = None)												
Mean Squared Error (MSE)	23.540	25.407	25.140	25.860	23.547	25.630	25.585	25.305	24.620	24.885	24.952	0.783
Root Mean Squared Error (RMSE)	4.852	5.041	5.014	5.085	4.853	5.063	5.058	5.030	4.962	4.988	4.995	0.079
Mean Absolute Error (MAE)	1.343	1.439	1.413	1.446	1.337	1.443	1.445	1.423	1.392	1.400	1.408	0.038
R ²	-0.501	-0.524	-0.564	-0.555	-0.516	-0.535	-0.633	-0.514	-0.483	-0.505	-0.533	0.041
Regression Tree (max_depth = 5)												
Mean Squared Error (MSE)	13.851	14.471	14.151	14.451	13.619	14.881	13.511	14.579	14.565	14.234	14.231	0.425
Root Mean Squared Error (RMSE)	3.722	3.804	3.762	3.801	3.690	3.858	3.676	3.818	3.816	3.773	3.772	0.056
Mean Absolute Error (MAE)	1.473	1.512	1.509	1.518	1.480	1.547	1.475	1.538	1.520	1.494	1.507	0.024
R ²	0.117	0.132	0.119	0.131	0.123	0.109	0.138	0.128	0.122	0.139	0.126	0.009
Random Forest (max_depth = None)												
Mean Squared Error (MSE)	12.149	12.588	12.510	12.752	12.087	13.110	11.964	12.894	12.580	12.276	12.491	0.352
Root Mean Squared Error (RMSE)	3.486	3.548	3.537	3.571	3.477	3.621	3.459	3.591	3.547	3.504	3.534	0.050
Mean Absolute Error (MAE)	1.333	1.365	1.365	1.383	1.343	1.412	1.348	1.386	1.367	1.350	1.365	0.022
R ²	0.226	0.245	0.222	0.233	0.222	0.215	0.236	0.229	0.242	0.258	0.233	0.012
Random Forest (max_depth = 5)												
Mean Squared Error (MSE)	13.648	14.290	13.954	14.278	13.382	14.646	13.412	14.406	14.262	14.057	14.033	0.407
Root Mean Squared Error (RMSE)	3.694	3.780	3.736	3.779	3.658	3.827	3.662	3.795	3.776	3.749	3.746	0.055
Mean Absolute Error (MAE)	1.467	1.505	1.499	1.516	1.467	1.537	1.471	1.521	1.507	1.487	1.498	0.023
R ²	0.130	0.143	0.132	0.142	0.138	0.123	0.144	0.138	0.141	0.150	0.138	0.007

2. [10 points] Analyze the 10-fold cross-validation results. Include any insightful observations. (At most ½ page)

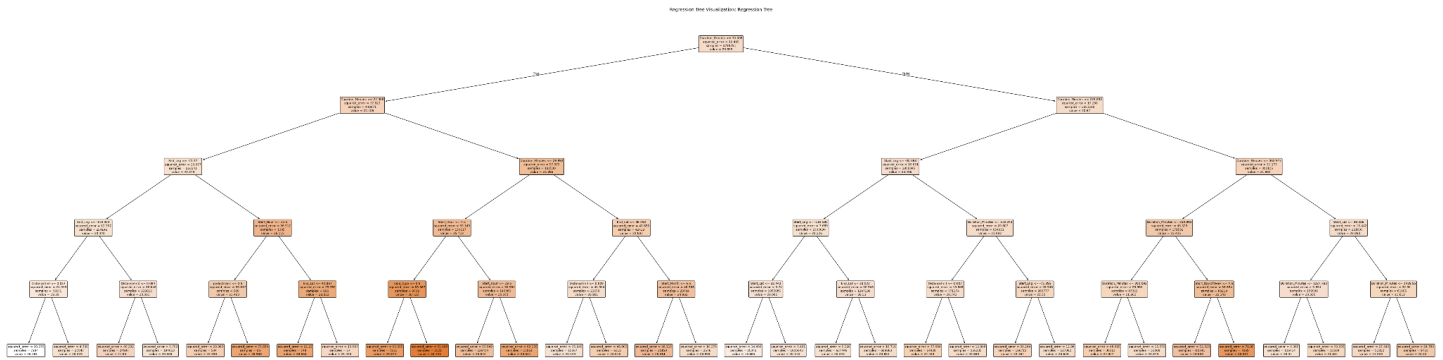
Part III.2 Final Regression Model and Analysis. (At most 2 pages)

- [10 Points] 3. What are the estimated generalization errors (MSE, RMSE, MAE, and R²) of this regression tree over the entire domain of car accidents? Explain your answer.

Metric	10-fold CV Mean	10-fold CV Std
MSE	14.367	0.088
RMSE	3.790	0.012
MAE	1.525	0.005
R ²	0.129	0.002

The errors are very stable. We can tell because the standard deviations for all metrics are very low. For example, R²'s standard deviation is 0.002. MAE's is 0.005, as another example. For these reasons, the estimates are precise and the performance of the regression tree does not depend on which accident data instances land in the test fold.

Regression Tree Depiction:



Qualitative Evaluation of the Regression Tree

Tree Size and Readability:

The tree is not too large, with a depth of 5 (not counting the root node). It has 63 nodes total, made up of 32 leaves and 31 internal nodes. Since the dataset is very large, every branch reaches the depth limit. The tree is easy to read and evaluate by hand because of its size (each path is at most five nodes/computations long). To help with interpreting the plot, nodes depicting higher predicted severity are darker than others.

3 Patterns Uncovered

1. Accidents cleared within 27-30 minute durations are more severe on average than any other durations. This is useful to domain experts as they could look into whether accidents handled within 30 minutes are often high-impact, since they'd likely need cleanup to arrive on-scene faster than less-severe accidents. Additionally, it is worth noting that duration is a result of an accident, so it can not help with preventing accidents of any severity in any way.
2. Nighttime accidents are the most severe. Within the 27-30 minute duration data instances, the most severe accidents were those that took place between 12am-4am. This makes sense, since people are more likely to drive at high speeds, disobey other traffic laws, and have lower visibility at night. This might also have to do with DUIs and drowsy-driving, which domain experts could look further into.
3. Amongst accidents with durations under 6 hours, weekend accidents are more severe than weekday accidents. There are many possible reasons as to why this pattern is being picked up, such as an increase in DUIs during the weekend due to an influx in partying/social events.

Overall Analysis of Regression Results

Predicting severity from these features is difficult. Even the best model, the unrestricted random forest, which has an R^2 of 0.233, leaves about 77% of variance in severity unexplained. The models in order of best to worst are as follows: Unrestricted random forest, random forest of depth 5, regression tree of depth 5, linear regression, and the dummy regressor. The final model for regression is a regression tree of depth 5 evaluated by 10-fold cross validation on all data instances (over 3.5 million). It yields an MSE of 14.37, MAE of 1.52, RMSE of 3.79, and an R^2 of 0.129. It gets within about 13% of the best forest while remaining readable, unlike trees of other sizes/experiments.

Parts IV and V. Clustering and Anomaly Detection [250 points] 1. [50 points] Summary of Experiments with K-means.

Pre-processing for all experiments: All experiments used the ‘advanced feature selection’ dataset we created.

Preprocessing steps included dropping features that weren’t meaningful, always had the same value, and couldn’t be encoded or transformed into a meaningful numerical value. We performed manual feature selection when many features represented the same aspect of the data (e.g., location). We also transformed certain features into new combined features when it made the data more meaningful and concise, such as creating the features ‘Distance’ and ‘Duration’. We grouped some features into more concise categories, such as weather, and encoded the remaining categorical variables as numerical values or dropped them if they were high-cardinality and low-value.

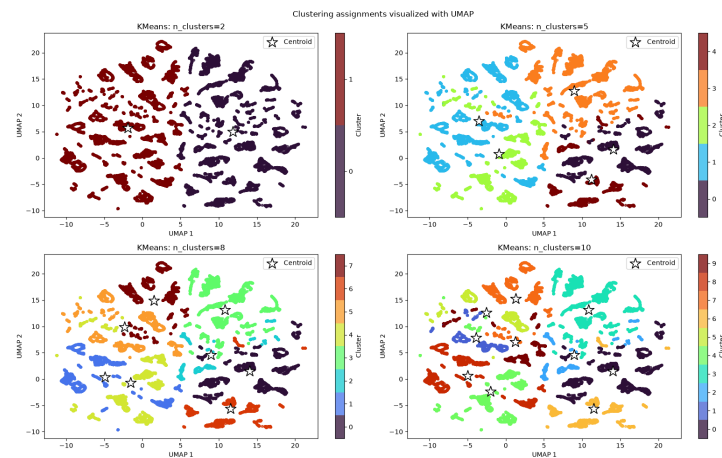
Finally, we used recursive feature elimination with cross-validation to select the top 14 features. For clustering, we used a stratified random sample of 1% of the data and normalized all values to 0-1. These preprocessing steps were necessary to meet sklearn data type requirements and optimize for available computing resources.

Experiments 1-4: Changing n_clusters with Euclidean distance function, iterations = auto

Experiment #	1, n_clusters = 2	2, n_clusters = 5	3, n_clusters = 8	4, n_clusters = 10
# clusters	2	5	8	10
# iterations	3	3	7	8
% of instances per cluster	{1: 51.2, 0: 48.8}	{1: 28.1, 3: 26.9, 2: 17.8, 0: 16.5, 4: 10.7}	{3: 18.7, 0: 16.4, 4: 14.0, 5: 12.9, 1: 12.9, 7: 11.4, 6: 8.7, 2: 4.9}	{3: 18.7, 0: 16.4, 8: 12.2, 4: 11.9, 7: 10.3, 6: 8.7, 9: 5.9, 5: 5.7, 1: 5.2, 2: 4.9}
SSE	159176.875000	72228.421875	39365.085938	33599.281250

Observations:

The results of this experiment are clear and easy to interpret in the UMAP diagram - the higher n_clusters is, the more color-coded clusters appear on the diagram. SSE decreases as the number of clusters increases, indicating that the centroid moves closer to the data points in its assigned clusters. However, this risks becoming overly granular and specific to the training data rather than the full dataset, so use the minimum n_clusters value needed to capture clustering trends in this dataset.



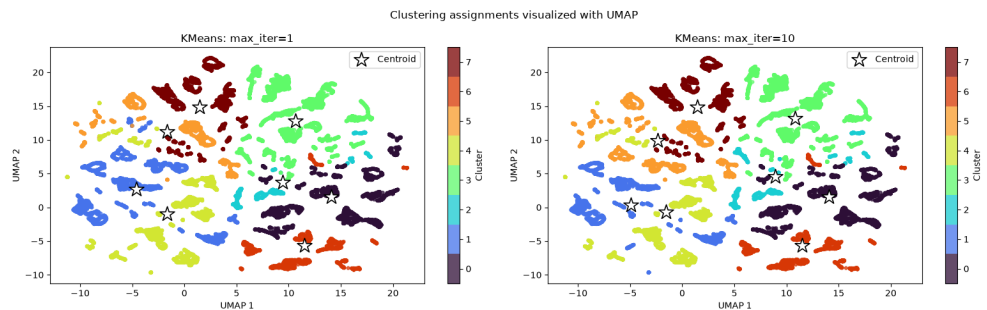
Experiments 5-6: Changing max_iter, with distance function Euclidean

Experiment #	5, max_iter = 1	6, max_iter = 10
# clusters	8	8
# iterations	1	10

% of instances per cluster	{3: 19.8, 1: 17.1, 0: 16.4, 4: 14.7, 7: 11.4, 6: 8.7, 5: 8.0, 2: 3.8}	{3: 18.7, 0: 16.4, 4: 14.0, 5: 12.9, 1: 12.9, 7: 11.4, 6: 8.7, 2: 4.9}
SSE	45448.179688	39365.078125

Observations:

When all other parameters are held constant, increasing max_iter from one appears to improve the overall quality of the clustering outcome. While clusters with max_iter=1 appear slightly more mixed and fragmented, clusters with max_iter=10 are more discrete overall. The SSE decreased by ~6000 from max_iter = 1 to max_iter = 10, indicating improved clustering quality with higher max_iter values.



Cluster feature similarity and differences:

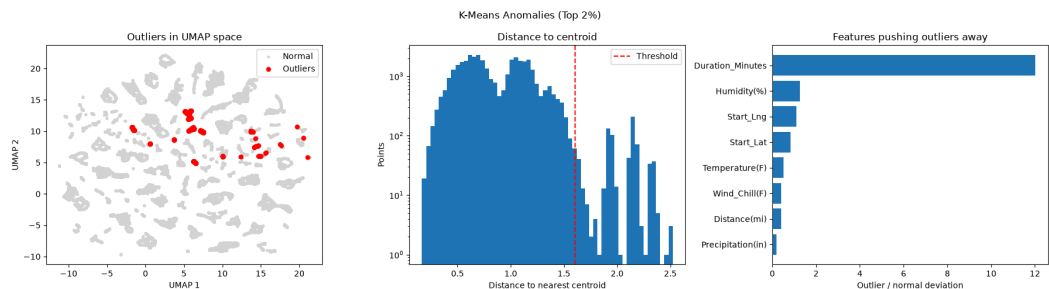
Feature	Variance
Start_Month	1.463080e+01
tod_lighting	9.217405e-01
Humidity(%)	5.292413e-03
...	...
Distance(mi)	1.886273e-07
Duration_Minutes	2.062069e-08
Precipitation(in)	2.894838e-09

For this report, we demonstrate KMeans cluster similarity and differences using experiment 4: n_clusters=10.

By analyzing between-group variance, we can see that month is the primary driver of cluster variance, followed by lighting. This means many clusters differ in month and lighting, while lower-ranked features with relatively low variance (such as precipitation, duration, and distance) are shared more commonly across multiple clusters in this analysis.

Anomaly detection: For this report, KMeans anomaly detection has been demonstrated using experiment 4: n_clusters=10. The anomaly score function $f(x)$ used in this code is the Euclidean distance from a data point to its assigned (closest) cluster centroid. The formula is $f(x) = ||x - C_i||$, where x is the data point, and C_i is the center of its assigned cluster.

We plotted the top 2% of K-Means Anomalies and found 760 (out of 37992 total points) were identified as outliers. Outliers are primarily driven by accident duration, as shown in the rightmost chart.



For duration, the deviation value is roughly 12, while every other feature is close to 1 or lower. This indicates that the duration of these anomalous events differs greatly from the other features.

[50 points] Summary of Experiments with Hierarchical Clustering (single link, complete link, average, Ward).

Pre-processing for all experiments:

All experiments used the ‘advanced feature selection’ dataset we created. Preprocessing steps included dropping features that weren’t meaningful, always had the same value, and couldn’t be encoded or transformed to a meaningful numerical value. We performed manual feature selection when many features represented the same aspect of the data (e.g., location). We also transformed certain features into new combined features when it made the data more meaningful and concise, such as creating the features ‘Distance’ and ‘Duration’. We grouped some features into more concise categories, such as weather, and encoded the remaining categorical variables as numerical values or dropped them if they were high-cardinality and low-value.

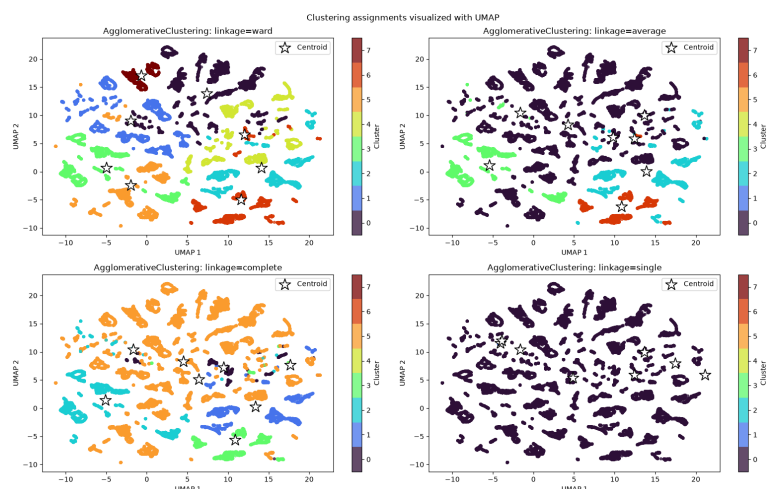
Finally, we used recursive feature elimination with cross-validation to select the top 14 features. For clustering, we used a stratified random sample of 1% of the data and normalized all values to 0-1. These preprocessing steps were necessary to meet sklearn data type requirements and optimize for available computing resources.

Experiments 1-4: Changing linkage for AgglomerativeClustering with n_clusters = 8, distance function Euclidean

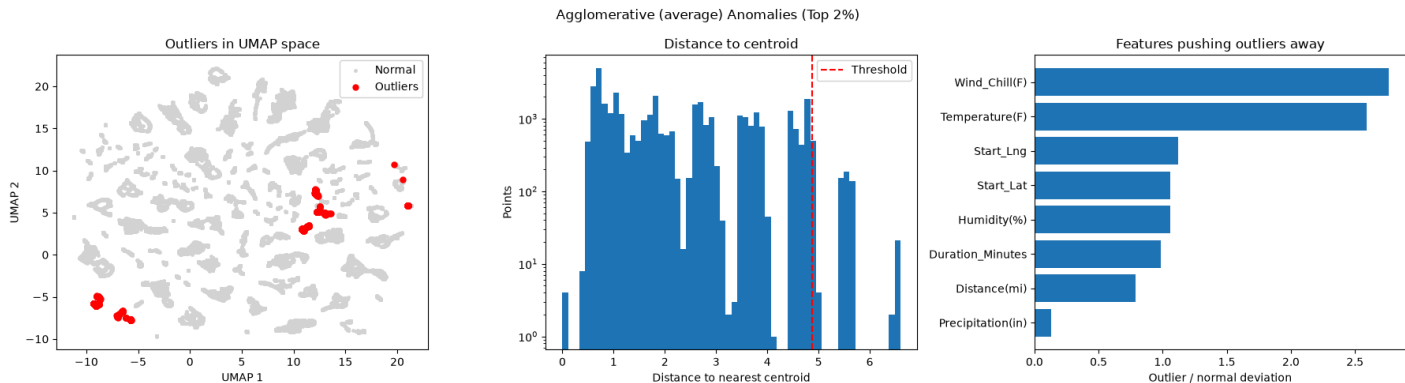
Experiment #	1, linkage = ward	2, linkage = average	3, linkage = complete	4, linkage = single
Link type	ward	average	complete	single
# clusters	8	8	8	8
% of instances per cluster	{0: 21.1, 1: 15.7, 4: 14.1, 3: 12.2, 5: 11.9, 2: 11.6, 6: 9.4, 7: 4.0}	{0: 67.8, 2: 13.0, 3: 12.0, 6: 7.2, 1: 0.0, 4: 0.0, 7: 0.0, 5: 0.0}	{5: 63.6, 2: 14.0, 1: 11.1, 3: 7.5, 0: 3.7, 7: 0.0, 4: 0.0, 6: 0.0}	{0: 99.6, 1: 0.3, 2: 0.0, 6: 0.0, 7: 0.0, 5: 0.0, 4: 0.0, 3: 0.0}
Time taken	30.694551	16.425420	23.534012	8.049288
Silhouette Score	0.3044	-0.1254	-0.0995	-0.5708

Observations:

The Ward linkage method is the only one that produced a balanced distribution of data points across clusters. Average, complete, and single are all very skewed, with >60% of data points in one cluster. This is visible in the UMAP clustering diagram to the right, where the colors are most balanced for Ward linkage. The silhouette score supports this analysis; Ward linkage has the highest silhouette score (0.3), while all other linkage types have sub-zero scores. However, the Ward method took the longest to run. This algorithm shows a sharp trade-off between time complexity and clustering accuracy.



Anomaly detection: For this report, hierarchical anomaly detection has been demonstrated using experiment 1: linkage = ward. The anomaly score function $f(x)$ used in this code is the Euclidean distance from a data point to its assigned (closest) cluster centroid. The formula is $f(x) = ||x - C_i||$, where x is the data point, and C_i is the center of its assigned cluster.



We plotted the top 2% of anomalies and found 760 (out of 37992 total points) were identified as outliers. In this model, wind chill primarily drives outliers, followed by temperature and, to a lesser extent, features such as lat/long, humidity, duration, and distance. Compared to k-means clustering, features contributing to outlier detection are much more balanced.

[50 points] Summary of Experiments with DBSCAN. At most 2 pages. *Include results of at least the best 5 experiments you ran.*

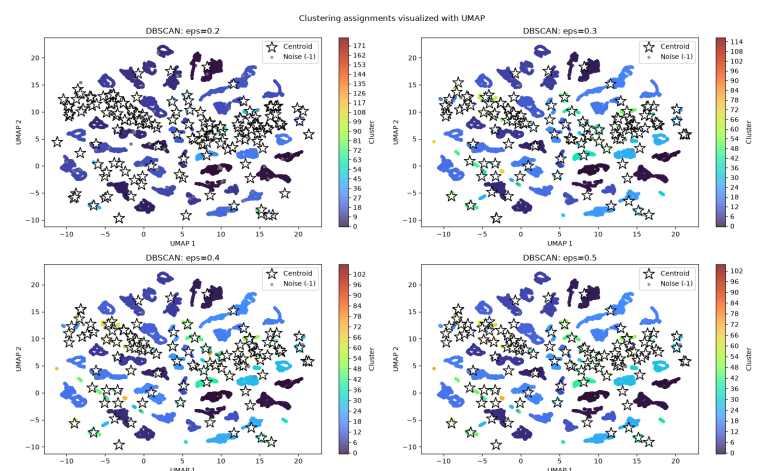
Pre-processing for all experiments: All experiments used the ‘advanced feature selection’ dataset we created. Preprocessing steps included dropping features that weren’t meaningful, always had the same value, and couldn’t be encoded or transformed to a meaningful numerical value. We performed manual feature selection when many features represented the same aspect of the data (e.g., location). We also transformed certain features into new combined features when it made the data more meaningful and concise, such as creating the features ‘Distance’ and ‘Duration’. We grouped some features into more concise categories, such as weather, and encoded the remaining categorical variables as numerical values or dropped them if they were high-cardinality and low-value.

Finally, we used recursive feature elimination with cross-validation to select the top 14 features. For clustering, we used a stratified random sample of 1% of the data and normalized all values to 0-1. These preprocessing steps were necessary to meet sklearn data type requirements and optimize for available computing resources.

Experiments 1-4: Different values of Epsilon

Experiment #	1, eps = 0.2	2, eps = 0.3	3, eps = 0.4	4, eps = 0.5
Epsilon	0.2	0.3	0.4	0.5
min_samples	5	5	5	5
# clusters	180	118	109	107
% of instances per cluster	{13: 5.4, 4: 4.9, 1: 4.8, 14: 4.6, 2: 4.5, 0: 4.5, 11: 4.5 ...}	{15: 5.4, 4: 4.9, 1: 4.8, 16: 4.6, 2: 4.5, ...}	{15: 5.4, 4: 4.9, 1: 4.8, 16: 4.6, 2: 4.5, 0: 4.5, ...}	{15: 5.4, 4: 4.9, 1: 4.8, 16: 4.6, 2: 4.5, 0: 4.5, ...}
Time taken	1.389515	1.226520	1.355192	1.489542

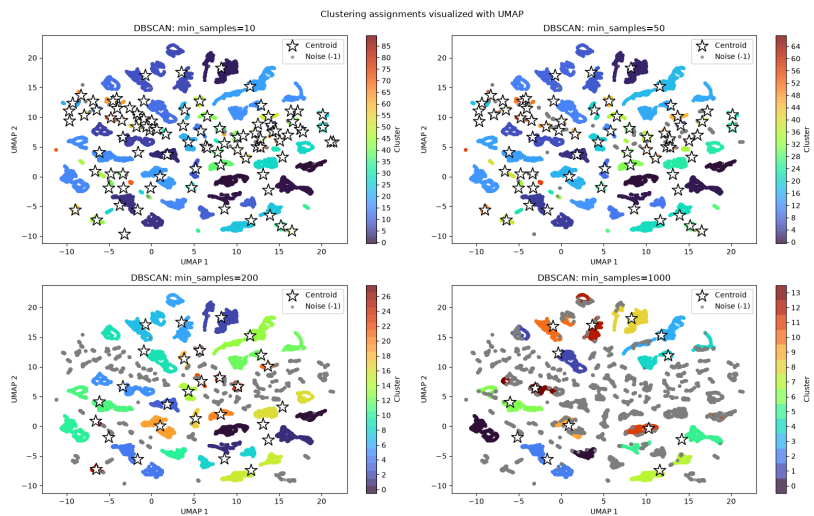
Observations: Using default parameters (other than Epsilon), this method identified over 100 clusters in all four experiments. However, the algorithm was very fast and took roughly the same amount of time across all four epsilon values. There appears to be a relationship between epsilon and the number of clusters, since higher epsilon values had lower cluster counts. We believe the default min_samples value is very high compared to the size of the input data, so it may have failed to identify noise appropriately. Our next experiment further addresses this issue and will identify better choices for the min_samples feature.



Experiments 5-8: Different values of min_samples

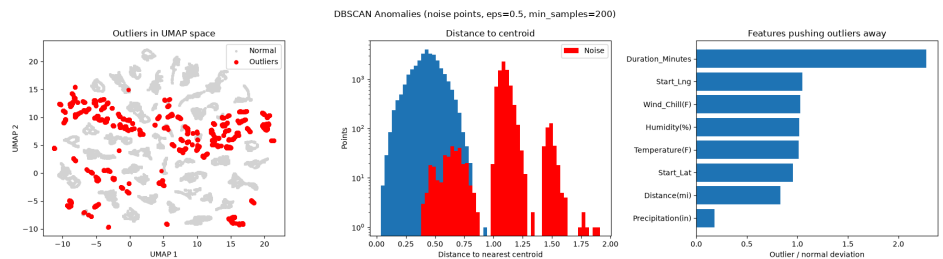
Experiment #	5, min_samples = 10	6, min_samples = 50	7, min_samples = 200	8, min_samples = 1000
Epsilon	0.5	0.5	0.5	0.5
min_samples	10	50	200	1000
# clusters	91	69	29	15
% of instances per cluster	{{15: 5.4, 4: 4.9, 1: 4.8, 16: 4.6, 2: 4.5, 0: 4.5, 12: 4.5, 17: 4.5, 19: 4.3 ...}}	{{13: 5.4, 4: 4.9, 1: 4.8, 14: 4.6, 2: 4.5, 0: 4.5, 11: 4.5, 15: 4.5, 17: 4.3 ...}}	{{-1: 20.8, 10: 5.4, 4: 4.9, 1: 4.8, 11: 4.6, 2: 4.5, 0: 4.5, 9: 4.5, 12: 4.5, 13: 4.3, 7: 4.0, 5: 4.0, ...}}	{{-1: 46.7, 0: 5.3, 2: 4.8, 5: 4.4, 8: 4.3, 6: 4.3, 4: 4.1, 1: 4.0, 3: 4.0, 7: 3.8, 10: 3.4, 12: 2.9, 11: 2.7, 9: 2.7, 13: 2.7}}
Time taken	1.672307	1.535932	1.476872	1.461446

Observations: min_samples seems critical when working with large datasets. Without an appropriate value for min_samples that is proportional to our dataset size, each ‘noise’ point quickly explodes into a large number of clusters. Min_samples = 10 identified 91 clusters, and as min_samples increases to 1000, the number of clusters decreases to 15. The run time also becomes marginally faster. The main risk of using a very high min_samples value is that clusters with too few samples can be discarded as ‘noise’, as seen in the diagram for min_samples = 1000, where a number of clusters were instead labeled as noise. We believe the best value in our diagram was min_samples = 200, which labels only the most dispersed points as noise while minimizing the number of clusters.



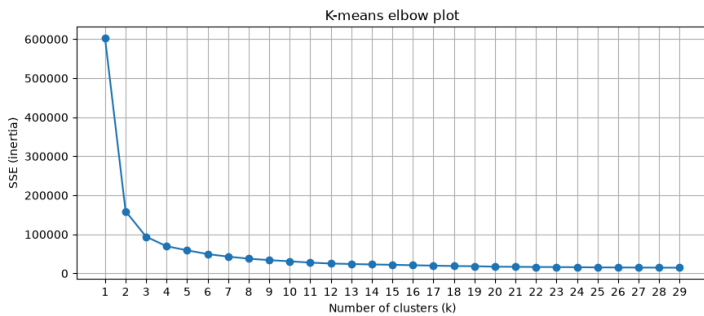
Anomaly detection: For this report, we demonstrated hierarchical anomaly detection using experiment 7: min_samples = 200. The anomaly score function $f(x)$ used in this code is the Euclidean distance from a data point to its assigned (closest) cluster centroid.

The formula is $f(x) = ||x - C_i||$, where x is the data point, and C_i is the center of its assigned cluster.



We plotted all of the “noise” and found 7909 (out of 37992 total points) were labeled as noise. In this model, duration primarily drives anomalies, followed by longitude, latitude, distance, wind chill, humidity, and temperature, which are all similarly important. Although many features contribute to anomaly detection in this model, duration is still the leading feature.

[50 points] Quantitative Analysis of Results and Discussion (at most 2 pages).

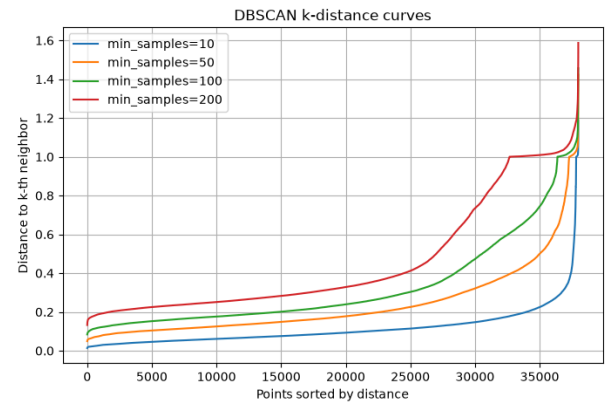


Obtaining good initial parameter values for K-means

To choose a good initial value for K-means, we created an elbow plot of the number of clusters versus SSE. Based on the elbow plot, we found that the default value for `n_clusters` ($k = 4$) was best, as it maximizes SSE without becoming too granular. After $k=4$, there were diminishing returns. We confirmed this by visually inspecting my experiments with different `n_clusters` values.

Obtaining good initial parameter values for DBSCAN

To choose a good initial epsilon value for DBSCAN, we plotted the k-distance curves. The elbow/knee point occurs at epsilon values between 0.2 and 0.4 depending on the minimum number of samples. We further tested different epsilon values and found that low values tended to be too granular. Min samples is also an important parameter - lower values increase granularity, while higher values tend to group multiple clusters together. We preferred the outcome at min samples 50 over 200 and 100, as it better preserved cluster separation.

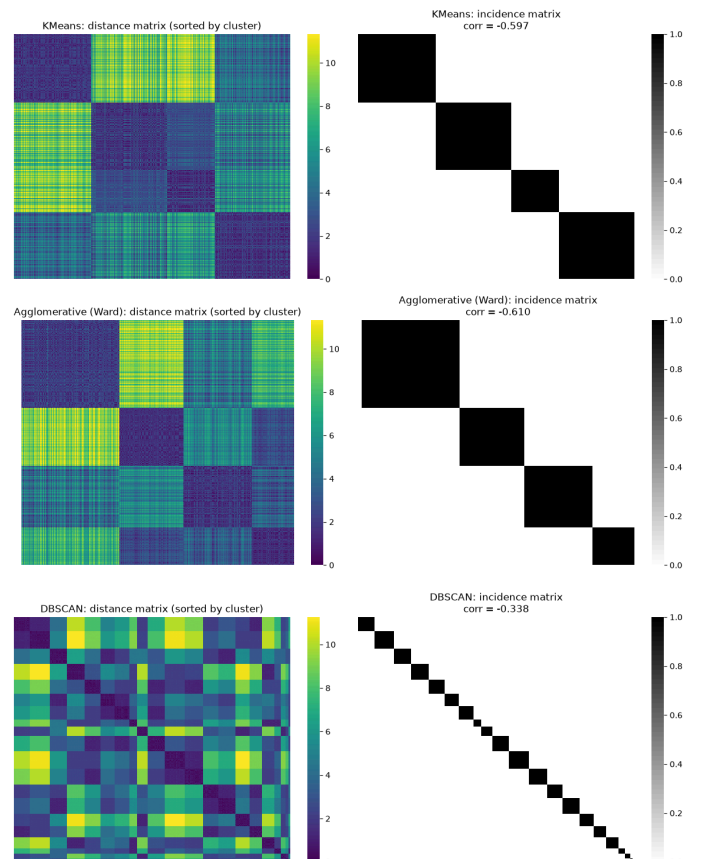


Based on all previous analysis, we have selected the following models for evaluation of clusterings:

```
KMeans(n_clusters=4, random_state=0,
max_iter=10).fit(X)
AgglomerativeClustering(linkage='ward',
connectivity=kneighbors_graph(X, n_neighbors=100,
include_self=False), n_clusters = 4).fit(X)
DBSCAN(eps=0.4, min_samples=200).fit(X)
```

Evaluation of clusterings

Internal indices: KMeans and Agglomerative (Ward) with 4 clusters performed similarly, achieving Silhouette scores around 0.397-0.398 and strong negative distance-incidence correlations (~ -0.60). In contrast, DBSCAN generated 21 clusters with 10,150 noise points ($\sim 26.7\%$ of the dataset), lower SSE (5,526.2), and yielded a higher Silhouette score (0.507) due to tight, dense core regions, though its distance-incidence correlation was weaker (-0.338).

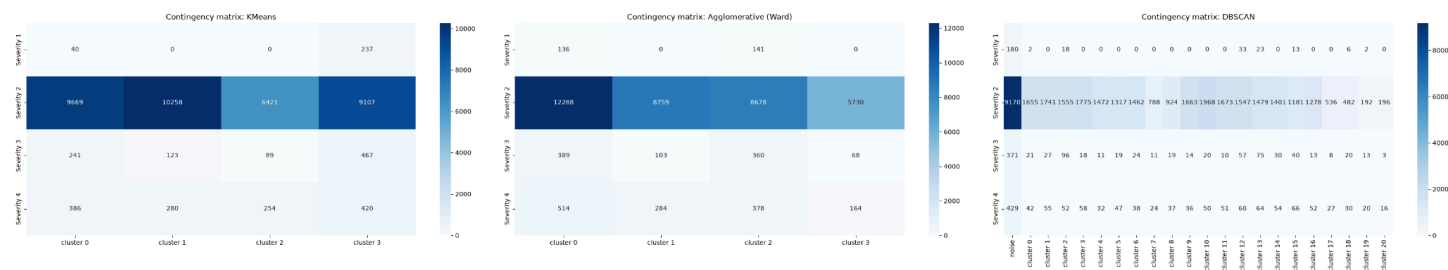


Method	n_clusters	n_noise	SSE	Silhouette	Dist-Incidence corr
KMeans	4	0	81261.625	0.3969	-0.5974
Agglomerative (Ward)	4	0	78558.039	0.3984	-0.6104
DBSCAN	21	10150	5526.242	0.5070	-0.3379

Relative indices: We found high alignment between KMeans (ARI = 0.531) and between KMeans and Agglomerative Ward (ARI = 0.481, NMI = 0.554), indicating similar clustering. This makes sense because they are both distance-based clustering methods. However, we found that DBSCAN shows low agreement with both KMeans and Agglomerative Ward (ARI ~ 0.14-0.15), probably because it utilized density-based clustering. This indicates it created significantly different groupings.

Clustering A	Clustering B	SSE A	SSE B	Adjusted Rand	NMI	AMI
KMeans (seed 0)	KMeans (seed 1)	81261.625	71015.000	0.5314	0.6192	0.6191
KMeans	Agglomerative (Ward)	81261.625	78558.039	0.4807	0.5535	0.5535
KMeans	DBSCAN	81261.625	5526.242	0.1480	0.4910	0.4908
Agglomerative (Ward)	DBSCAN	78558.039	5526.242	0.1418	0.4864	0.4862

External indices: We found that these had very low metrics across all three algorithms, with homogeneity peaking at 0.0384 (DBSCAN) and V-measures below 0.015. This indicates that the feature clusters discovered by unsupervised methods did not align with the severity feature.

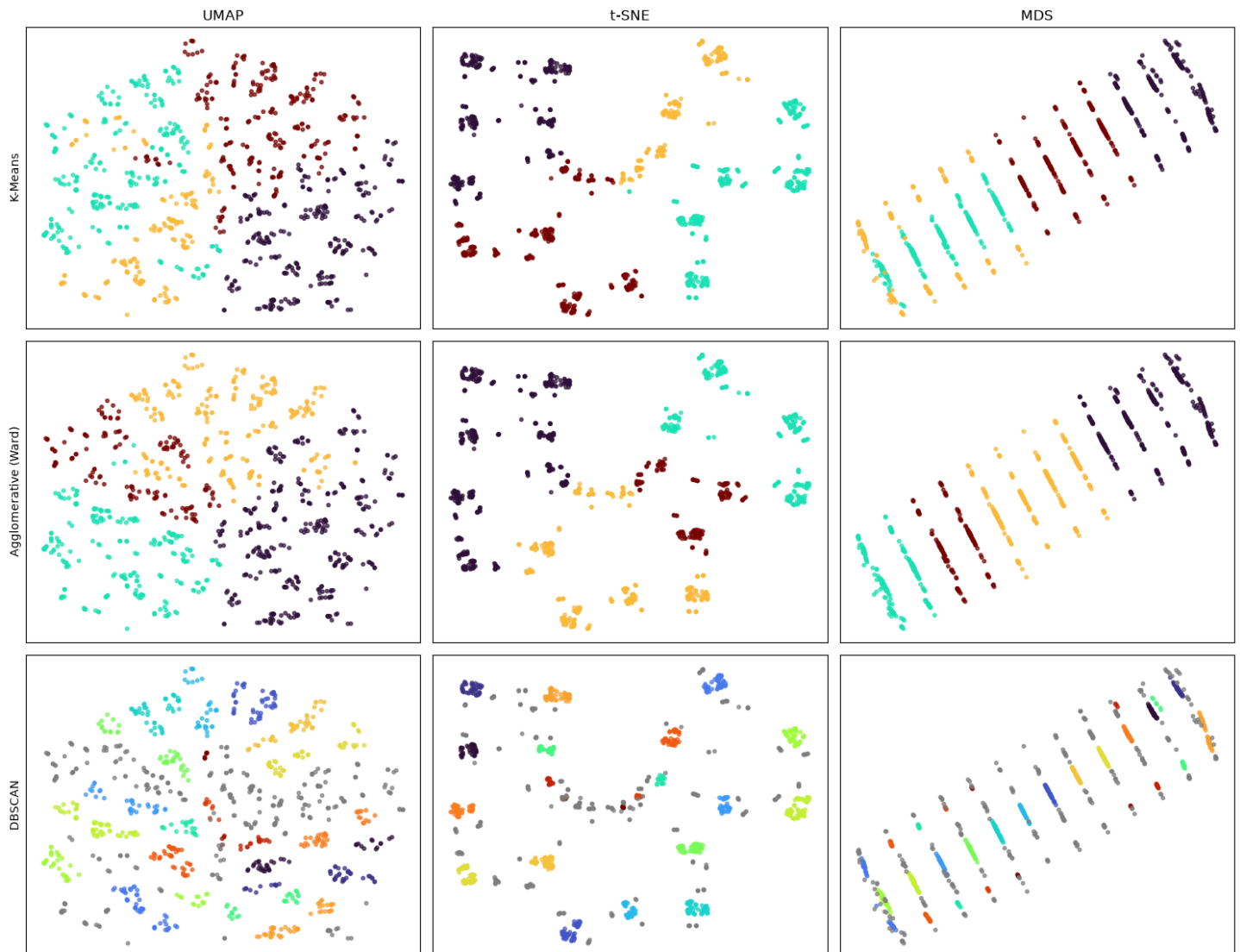


Method	Homogeneity	Completeness	V-measure
KMeans	0.0363	0.0082	0.0133
Agglomerative (Ward)	0.0229	0.0052	0.0085
DBSCAN	0.0384	0.0043	0.0078

[50 points] Qualitative Analysis of Results and Visualizations (at most 2 pages)

Visualizations of the clustering

Selected clusterings across UMAP, t-SNE, and MDS



Colors denote cluster assignments within each panel; gray points are DBSCAN noise. Same 1,000-accident subset in every projection.

Inspection of the actual clusters' members (table in ipynb file)

All four kmeans clusters have similar median accident distances and median durations. The clusters are most different in location and weather: for example, Cluster 0 has a median temperature of 49 degrees F, humidity of 69%, and location (36.24, -91.03); Cluster 3 has 73 degrees F, 59%, and (36.16, -86.31). In the four agglomerative clusters, the cooler clusters have median temperatures of 51-52 degrees F, compared with 71-78 degrees F in its warmer clusters. Their median accident distances remain similar (0.25-0.28 mi), suggesting that weather and geography are more impactful than accident distance.

With DBSCAN, we identified 21 smaller clusters and labeled 10150 accidents as noise. Its groups are most distinguished by geography and weather. For example, DBSCAN Clusters 19 and 20 both lie around (34-35, 81-83) and have median distances of 0.26-0.28 min, while their median temperatures differ (60 degrees F vs. 75 degrees F). Cluster sizes range from 215 to 2038 accidents.

Outlier commonalities

	Approach A	Approach B	Shared accidents	Jaccard overlap
0	K-Means top 2% distance	Agglomerative top 2% distance	488	0.473
1	K-Means top 2% distance	DBSCAN noise	545	0.053
2	Agglomerative top 2% distance	DBSCAN noise	545	0.053

We found that K-Means and Agglomerative clustering have the most shared outliers. Each flags 760 accidents as its top 2% most distant from cluster centers and 488 are shared (Jaccard overlap 0.473). These outliers were usually warmer and farther west than the rest of the data, with median temps of 70-73 degrees F versus 62 degrees F, and median latitudes are about 36.8 versus 36.15.

Of those distance-based outliers, 545 also appear as DBSCAN noise. That is about 72% of each 760-case set, but only 5.3% Jaccard overlap with DBSCAN's much larger noise set of 10150 accidents. All three flag at least 273 shared accidents. DBSCAN noise does not show the same weather pattern as the K-mean and Agglomerative clusterin. Its median temperature is 62 degrees F versus 63 degrees F for non-noise samples. This makes sense, because we expected that the two distanced based methods would have a more similar subset while the density-based DBSCANN would have different results.

Additional analysis from POV dataset domain

- 1) We found that the severity is not a strong separator. Severity 2 is the majority in every selected cluster, generally about 85-97%. The clusters should not be interpreted as having any real correlation with severity.
- 2) We did, however, find that weather and temperature distinguish some groups. For example, K-Means clusters have similar median accident distances (0.25-0.29 mi), but their median temperatures range from 49 degrees F to 73 degrees F and their median coordinates differ. As a result, we believe that regional or seasonal context may matter more than distance for these clusters. This could represent accident clusters where accidents were caused by driving in more extreme temperature and weather conditions.
- 3) Duration is strongly right-skewed. Cluster median durations are mostly around 88-113 minutes, while several means are hundreds or even thousands of minutes. A small number of accidents had the longest duration.

PART VI: [30 points] Reflection. *This part can be completed individually or as a group; your choice.* Answer the following questions summarizing your highlights for this project (which we'll use for the project's in-class discussion):

- Reflect on the content of (i.e., the course material covered in) this project. Use "big picture" thinking when answering these questions:
 - a) What are the top 3 things you learned about data mining from working on this project?
 1. The data is significantly larger than any of us have worked on before. Adapting to using such a large dataset was rather difficult, but we managed anyway.
 2. In addition to managing a large number of records, there was a very large amount of attributes to sort through. We had to use some rather severe techniques in order to bring the data down to a manageable level.
 3. Trying to manage all of the different types of techniques for regression, classification, and clustering has been difficult. Trying to understand and identify the proper inputs to use has been interesting.
 - b) What are the top 3 things you learned about the data domain (i.e., car accidents) from this project?
 1. The distance that a crash takes place over is one of the most significant aspects of how severe it is.
 - 2.
 - c) What are the top 3 things you want to learn more about the data mining topics covered by this project?
 1. What is the best way to manage such a large data
 2. How big of a sample is to be considered reasonable, and how would we decided that for vary sizes of databases
 3. What methods would help us speed up our experimentation with various attributes of the decision trees, random forests, or other methods.
- Reflect now on how you worked on this project:
 - d) If you could start working on this project again, what would you do differently?

Our number one concern was trying to manage our time more effectively. We spend a significant amount of time trying to manage the significantly oversized dataset. If we had instead gone straight to managing the size of the data, instead of trying to work with it, we would have had a much easier time in applying our techniques.

Start earlier, instead of trying to work with the whole dataset, only work with subsets, apply my rigorous preprocessing immediately

Authorship: Although each student on the team is expected to be involved in every aspect of the project, describe in detail here the main contributions that each of the team members made to this project. This authorship description must accurately reflect the work done by each team member, and must be approved by all of the members of the team (at most 1/3 page)

We arranged several meetings to go over what exactly is expected of us, as well as to collaboratively work on our parts. We did our best to divide the work evenly. We made sure to divide the different sections into their constituent parts, with Tucker taking over the Classification portion, Charlotte taking care of the regression, and Sarah handling the clustering and Anomaly detection portion. In order to ensure that the work distribution is fair, Tucker and Charlotte handled the non-specific writeups, such as the introduction and the reflection.

Classification Tables:

Dummy:

Dummy Classifier	Fold 1				Severity Value	Precision	Recall	F1
0	2768	0	0	Accuracy 0.933	0	0	0	0
0	354553	0	0		1	0.933	1	0.965
0	9203	0	0		2	0	0	0
0	13402	0	0		3	0	0	0
Dummy Classifier	Fold 2							
0	2767	0	0	Accuracy 0.933	0	0	0	0
0	354553	0	0		1	0.933	1	0.965
0	9203	0	0		2	0	0	0
0	13402	0	0		3	0	0	0
Dummy Classifier	Fold 3							
0	2767	0	0	Accuracy 0.933	0	0	0	0
0	354553	0	0		1	0.933	1	0.965
0	9203	0	0		2	0	0	0
0	13402	0	0		3	0	0	0
Dummy Classifier	Fold 4							

0	2767	0	0	Accuracy 0.933	0	0	0	0
0	354553	0	0		1	0.933	1	0.965
0	9203	0	0		2	0	0	0
0	13402	0	0		3	0	0	0
Dummy Classifier	Fold 5							
0	2767	0	0	Accuracy 0.933	0	0	0	0
0	354553	0	0		1	0.933	1	0.965
0	9203	0	0		2	0	0	0
0	13402	0	0		3	0	0	0
Dummy Classifier	Fold 6							
0	2767	0	0	Accuracy: 0.933	0	0	0	0
0	354554	0	0		1	0.933	1	0.965
0	9202	0	0		2	0	0	0
0	13402	0	0		3	0	0	0
Dummy Classifier	Fold 7							
0	2767	0	0	Accuracy: 0.933	0	0	0	0
0	354554	0	0		1	0.933	1	0.965
0	9202	0	0		2	0	0	0
0	13402	0	0		3	0	0	0

Dummy Classifier	Fold 8							
0	2767	0	0	Accuracy: 0.933	0	0	0	0
0	354554	0	0		1	0.933	1	0.965
0	9202	0	0		2	0	0	0
0	13402	0	0		3	0	0	0
Dummy Classifier	Fold 9							
0	2767	0	0	Accuracy: 0.933	0	0	0	0
0	354554	0	0		1	0.933	1	0.965
0	9202	0	0		2	0	0	0
0	13402	0	0		3	0	0	0
Dummy Classifier	Fold 10							
0	2767	0	0	Accuracy: 0.933	0	0	0	0
0	354554	0	0		1	0.933	1	0.965
0	9202	0	0		2	0	0	0
0	13402	0	0		3	0	0	0

max_depth=None Decision Tree:

fold 1							
2048	450	195	75				

510	341778	3512	8753				
192	3123	5144	744				
48	8455	715	4184				
Accuracy	0.93		Precision		Recall		F1
Severity	0		0.732		0.74		0.736
Severity	1		0.966		0.964		0.965
Severity	2		0.538		0.559		0.548
Severity	3		0.304		0.312		0.308
fold 2							
	2039	476	181	71			
	481	341835	3508	8729			
	141	3101	5200	761			
	54	8425	685	4238			
Accuracy	0.93		Precision		Recall		F1
Severity	0		0.751		0.737		0.744
Severity	1		0.966		0.964		0.965
Severity	2		0.543		0.565		0.554
Severity	3		0.307		0.316		0.312
fold 3							
	2048	495	163	61			
	476	342061	3368	8648			
	181	3113	5157	752			

	54	8399	702	4247			
Accuracy	0.93		Precision		Recall		F1
Severity	0		0.742		0.74		0.741
Severity	1		0.966		0.965		0.965
Severity	2		0.549		0.56		0.555
Severity	3		0.31		0.317		0.313
fold 4							
	2081	452	181	53			
	506	341652	3459	8936			
	185	3206	5081	731			
	55	8339	697	4311			
Accuracy	0.929						
Severity	0 Precision	0.736	Recall	0.752	F1		
Severity	1 Precision	0.966	Recall	0.964	F1		
Severity	2 Precision	0.539	Recall	0.552	F1		
Severity	3 Precision	0.307	Recall	0.322	F1		
fold 5							
	2104	464	141	58			
	568	341640	3420	8925			
	189	3170	5112	732			
	67	8402	656	4277			
Accuracy	0.929		Precision		Recall		F1

Severity	0		0.719		0.76		0.739
Severity	1		0.966		0.964		0.965
Severity	2		0.548		0.555		0.552
Severity	3		0.306		0.319		0.312
fold 6							
	1996	521	192	58			
	504	341888	3389	8773			
	169	3052	5205	776			
	55	8338	691	4318			
Accuracy	0.93		Precision		Recall		F1
Severity	0		0.733		0.721		0.727
Severity	1		0.966		0.964		0.965
Severity	2		0.549		0.566		0.557
Severity	3		0.31		0.322		0.316
fold 7							
	2012	512	178	65			
	526	341721	3459	8848			
	156	3099	5187	760			
	45	8382	703	4272			
Accuracy	0.93		Precision		Recall		F1
Severity	0		0.735		0.727		0.731
Severity	1		0.966		0.964		0.965

Severity	2		0.544		0.564		0.554
Severity	3		0.306		0.319		0.312
fold 8							
	2017	521	183	46			
	513	341638	3493	8910			
	192	3065	5228	717			
	61	8407	678	4256			
Accuracy	0.929		Precision		Recall		F1
Severity	0		0.725		0.729		0.727
Severity	1		0.966		0.964		0.965
Severity	2		0.546		0.568		0.557
Severity	3		0.306		0.318		0.311
fold 9							
	2046	494	170	57			
	503	341860	3463	8728			
	186	3132	5138	746			
	50	8521	712	4119			
Accuracy	0.93		Precision		Recall		F1
Severity	0		0.735		0.739		0.737
Severity	1		0.966		0.964		0.965
Severity	2		0.542		0.558		0.55
Severity	3		0.302		0.307		0.305

fold 10							
	2081	445	190	51			
	517	341780	3357	8900			
	171	3070	5180	781			
	63	8523	679	4137			
Accuracy	0.93		Precision		Recall		F1
Severity	0		0.735		0.752		0.743
Severity	1		0.966		0.964		0.965
Severity	2		0.551		0.563		0.557
Severity	3		0.298		0.309		0.303

max_depth=5 Decision Tree

fold 1							
Confusion Matrix							
	343	191	21	0			
	113	70403	240	203			
	30	1159	558	74			
	13	2257	99	282			
Accuracy	0.942						
Severity	0	Precision	0.687	Recall	0.618	F1	0.651
Severity	1	Precision	0.951	Recall	0.992	F1	0.971
Severity	2	Precision	0.608	Recall	0.306	F1	0.407

Severity	3	Precision	0.504	Recall	0.106	F1	0.176
fold 2							
Confusion Matrix							
	325	207	19	3			
	60	70452	238	208			
	25	1130	574	92			
	26	2267	73	286			
Accuracy	0.943						
Severity	0	Precision	0.745	Recall	0.587	F1	0.657
Severity	1	Precision	0.951	Recall	0.993	F1	0.972
Severity	2	Precision	0.635	Recall	0.315	F1	0.421
Severity	3	Precision	0.486	Recall	0.108	F1	0.176
fold 3							
Confusion Matrix							
	331	199	24	0			
	99	70416	270	173			
	33	1155	569	64			
	15	2293	89	255			
Accuracy	0.942						
Severity	0 Precision	0.692	Recall	0.597	F1	0.641	
Severity	1 Precision	0.951	Recall	0.992	F1	0.971	
Severity	2 Precision	0.598	Recall	0.312	F1	0.41	
Severity	3 Precision	0.518	Recall	0.096	F1	0.162	

fold 4							
Confusion Matrix							
	309	224	18	3			
	53	70513	218	174			
	26	1167	572	56			
	13	2286	88	265			
Accuracy	0.943						
Severity	0	Precision	0.771	Recall	0.558	F1	0.647
Severity	1	Precision	0.95	Recall	0.994	F1	0.972
Severity	2	Precision	0.638	Recall	0.314	F1	0.421
Severity	3	Precision	0.532	Recall	0.1	F1	0.168
fold 5							
Confusion Matrix							
	340	190	24	1			
	71	70478	243	166			
	42	1101	620	57			
	21	2286	87	258			
Accuracy	0.944						
Severity	0	Precision	0.717	Recall	0.613	F1	0.661
Severity	1	Precision	0.952	Recall	0.993	F1	0.972
Severity	2	Precision	0.637	Recall	0.341	F1	0.444
Severity	3	Precision	0.535	Recall	0.097	F1	0.165
fold 6							

Confusion Matrix							
	331	193	25	6			
	51	70484	263	160			
	30	1138	590	63			
	16	2287	80	268			
Accuracy	0.943						
Severity	0	Precision	0.773	Recall	0.596	F1	0.673
Severity	1	Precision	0.951	Recall	0.993	F1	0.972
Severity	2	Precision	0.616	Recall	0.324	F1	0.425
Severity	3	Precision	0.539	Recall	0.101	F1	0.17
fold 7							
Confusion Matrix							
	331	200	22	2			
	68	70465	250	175			
	45	1150	560	66			
	20	2292	100	239			
Accuracy	0.942						
Severity	0	Precision	0.713	Recall	0.596	F1	0.65
Severity	1	Precision	0.951	Recall	0.993	F1	0.971
Severity	2	Precision	0.601	Recall	0.308	F1	0.407
Severity	3	Precision	0.496	Recall	0.09	F1	0.153
fold 8							

Confusion Matrix							
	335	197	23	0			
	62	70539	178	179			
	33	1186	542	60			
	19	2295	72	265			
Accuracy	0.943						
Severity	0	Precision	0.746	Recall	0.604	F1	0.667
Severity	1	Precision	0.95	Recall	0.994	F1	0.972
Severity	2	Precision	0.665	Recall	0.298	F1	0.411
Severity	3	Precision	0.526	Recall	0.1	F1	0.168
fold 9							
Confusion Matrix							
	341	179	31	4			
	63	70421	267	207			
	21	1150	560	90			
	24	2263	68	296			
Accuracy	0.943						
Severity	0	Precision	0.759	Recall	0.614	F1	0.679
Severity	1	Precision	0.951	Recall	0.992	F1	0.972
Severity	2	Precision	0.605	Recall	0.308	F1	0.408
Severity	3	Precision	0.496	Recall	0.112	F1	0.182
fold 10							

Confusion Matrix							
	339	192	17	7			
	80	70459	206	213			
	31	1120	600	70			
	14	2272	77	288			
Accuracy	0.943						
Severity	0	Precision	0.731	Recall	0.611	F1	0.665
Severity	1	Precision	0.952	Recall	0.993	F1	0.972
Severity	2	Precision	0.667	Recall	0.329	F1	0.441
Severity	3	Precision	0.498	Recall	0.109	F1	0.178

max_depth=None Random Forest

fold 1							
Confusion Matrix							
	58	78	2	0			
	4	17731	9	0			
	0	400	55	0			
	3	657	0	0			
Accuracy	0.939						
Severity	0	Precision	0.892	Recall	0.42	F1	0.571
Severity	1	Precision	0.94	Recall	0.999	F1	0.969
Severity	2	Precision	0.833	Recall	0.121	F1	0.211

Severity	3	Precision	0	Recall	0	F1	0
fold 2							
Confusion Matrix							
	57	79	2	0			
	3	17734	7	0			
	3	410	42	0			
	2	653	5	0			
Accuracy	0.939						
Severity	0	Precision	0.877	Recall	0.413	F1	0.562
Severity	1	Precision	0.939	Recall	0.999	F1	0.969
Severity	2	Precision	0.75	Recall	0.092	F1	0.164
Severity	3	Precision	0	Recall	0	F1	0
fold 3							
Confusion Matrix							
	55	81	2	0			
	6	17730	8	0			
	9	400	47	0			
	5	652	2	0			
Accuracy	0.939						
Severity	0	Precision	0.733	Recall	0.399	F1	0.516
Severity	1	Precision	0.94	Recall	0.999	F1	0.969
Severity	2	Precision	0.797	Recall	0.103	F1	0.183
Severity	3	Precision	0	Recall	0	F1	0

fold 4							
Confusion Matrix							
	54	82	1	0			
	2	17739	3	0			
	3	410	43	0			
	2	655	2	0			
Accuracy	0.939						
Severity	0	Precision	0.885	Recall	0.394	F1	0.545
Severity	1	Precision	0.939	Recall	1	F1	0.969
Severity	2	Precision	0.878	Recall	0.094	F1	0.17
Severity	3	Precision	0	Recall	0	F1	0
fold 5							
Confusion Matrix							
	50	85	2	0			
	0	17735	8	0			
	3	405	48	0			
	2	658	0	0			
Accuracy	0.939						
Severity	0	Precision	0.909	Recall	0.365	F1	0.521
Severity	1	Precision	0.939	Recall	1	F1	0.968
Severity	2	Precision	0.828	Recall	0.105	F1	0.187
Severity	3	Precision	0	Recall	0	F1	0
fold 6							

Confusion Matrix							
	53	84	0	0			
	4	17731	8	0			
	6	398	52	0			
	0	658	2	0			
Accuracy	0.939						
Severity	0	Precision	0.841	Recall	0.387	F1	0.53
Severity	1	Precision	0.94	Recall	0.999	F1	0.969
Severity	2	Precision	0.839	Recall	0.114	F1	0.201
Severity	3	Precision	0	Recall	0	F1	0
fold 7							
Confusion Matrix							
	42	95	0	0			
	2	17735	6	0			
	5	404	47	0			
	0	659	1	0			
Accuracy	0.938						
Severity	0	Precision	0.857	Recall	0.307	F1	0.452
Severity	1	Precision	0.939	Recall	1	F1	0.968
Severity	2	Precision	0.87	Recall	0.103	F1	0.184
Severity	3	Precision	0	Recall	0	F1	0
fold 8							

Confusion Matrix							
	60	78	0	0			
	6	17734	3	0			
	4	410	41	0			
	3	655	2	0			
Accuracy	0.939						
Severity	0	Precision	0.822	Recall	0.435	F1	0.569
Severity	1	Precision	0.939	Recall	0.999	F1	0.969
Severity	2	Precision	0.891	Recall	0.09	F1	0.164
Severity	3	Precision	0	Recall	0	F1	0
fold 9							
Confusion Matrix							
	47	89	2	0			
	8	17731	4	0			
	2	410	43	0			
	6	652	2	0			
Accuracy	0.938						
Severity	0	Precision	0.746	Recall	0.341	F1	0.468
Severity	1	Precision	0.939	Recall	0.999	F1	0.968
Severity	2	Precision	0.843	Recall	0.095	F1	0.17
Severity	3	Precision	0	Recall	0	F1	0
fold 10							

Confusion Matrix							
	61	76	1	0			
	2	17734	7	0			
	2	408	45	0			
	2	657	1	0			
Accuracy	0.939						
Severity	0	Precision	0.91	Recall	0.442	F1	0.595
Severity	1	Precision	0.94	Recall	0.999	F1	0.969
Severity	2	Precision	0.833	Recall	0.099	F1	0.177
Severity	3	Precision	0	Recall	0	F1	0

max_depth=5 Random Forest

fold 1							
Confusion Matrix							
	4	134	0	0			
	0	17744	0	0			
	0	455	0	0			
	0	660	0	0			
Accuracy	0.934						
Severity	0	Precision	1	Recall	0.029	F1	0.056
Severity	1	Precision	0.934	Recall	1	F1	0.966
Severity	2	Precision	0	Recall	0	F1	0

Severity	3	Precision	0	Recall	0	F1	0
fold 2							
Confusion Matrix							
	1	137	0	0			
	0	17744	0	0			
	0	455	0	0			
	0	660	0	0			
Accuracy	0.934						
Severity	0	Precision	1	Recall	0.007	F1	0.014
Severity	1	Precision	0.934	Recall	1	F1	0.966
Severity	2	Precision	0	Recall	0	F1	0
Severity	3	Precision	0	Recall	0	F1	0
fold 3							
Confusion Matrix							
	6	132	0	0			
	0	17744	0	0			
	0	456	0	0			
	0	659	0	0			
Accuracy	0.934						
Severity	0	Precision	1	Recall	0.043	F1	0.083
Severity	1	Precision	0.934	Recall	1	F1	0.966
Severity	2	Precision	0	Recall	0	F1	0
Severity	3	Precision	0	Recall	0	F1	0

fold 4							
Confusion Matrix							
	1	136	0	0			
	0	17744	0	0			
	0	455	1	0			
	0	659	0	0			
Accuracy	0.934						
Severity	0	Precision	1	Recall	0.007	F1	0.014
Severity	1	Precision	0.934	Recall	1	F1	0.966
Severity	2	Precision	1	Recall	0.002	F1	0.004
Severity	3	Precision	0	Recall	0	F1	0
fold 5							
Confusion Matrix							
	1	136	0	0			
	0	17743	0	0			
	0	456	0	0			
	0	660	0	0			
Accuracy	0.934						
Severity	0	Precision	1	Recall	0.007	F1	0.014
Severity	1	Precision	0.934	Recall	1	F1	0.966
Severity	2	Precision	0	Recall	0	F1	0
Severity	3	Precision	0	Recall	0	F1	0
fold 6							

Confusion Matrix							
	9	128	0	0			
	0	17743	0	0			
	0	456	0	0			
	0	660	0	0			
Accuracy	0.935						
Severity	0	Precision	1	Recall	0.066	F1	0.123
Severity	1	Precision	0.934	Recall	1	F1	0.966
Severity	2	Precision	0	Recall	0	F1	0
Severity	3	Precision	0	Recall	0	F1	0
fold 7							
Confusion Matrix							
	2	135	0	0			
	0	17743	0	0			
	0	456	0	0			
	0	660	0	0			
Accuracy	0.934						
Severity	0	Precision	1	Recall	0.015	F1	0.029
Severity	1	Precision	0.934	Recall	1	F1	0.966
Severity	2	Precision	0	Recall	0	F1	0
Severity	3	Precision	0	Recall	0	F1	0
fold 8							

Confusion Matrix							
	5	133	0	0			
	0	17743	0	0			
	0	455	0	0			
	0	660	0	0			
Accuracy	0.934						
Severity	0	Precision	1	Recall	0.036	F1	0.07
Severity	1	Precision	0.934	Recall	1	F1	0.966
Severity	2	Precision	0	Recall	0	F1	0
Severity	3	Precision	0	Recall	0	F1	0
fold 9							
Confusion Matrix							
	3	135	0	0			
	0	17743	0	0			
	0	455	0	0			
	0	660	0	0			
Accuracy	0.934						
Severity	0	Precision	1	Recall	0.022	F1	0.043
Severity	1	Precision	0.934	Recall	1	F1	0.966
Severity	2	Precision	0	Recall	0	F1	0
Severity	3	Precision	0	Recall	0	F1	0
fold 10							

Confusion Matrix							
	5	133	0	0			
	0	17743	0	0			
	0	455	0	0			
	0	660	0	0			
Accuracy	0.934						
Severity	0	Precision	1	Recall	0.036	F1	0.07
Severity	1	Precision	0.934	Recall	1	F1	0.966
Severity	2	Precision	0	Recall	0	F1	0
Severity	3	Precision	0	Recall	0	F1	0

Bro why